

Technological Innovations in Agriculture: A Philosophy and Sociology of Science Approach

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ABSTRACT ORIGINAL

How do human interventions into the environment motivated by different aims transform agriculture in ways that create new causal relationships between organisms above and below ground? We provide a conceptual framework for a philosophical and sociological approach to agricultural biotechnology and its multiple impacts on agricultural systems. We begin with a brief account of the history of the philosophy of the agricultural sciences and the early reluctance of philosophers of science to engage in the philosophy of the applied sciences. Applied sciences in general and the agricultural sciences in particular rely on contingencies in social practice. Here, one would expect diversity among different disciplines and goals. Instead of the basic sciences' aim of discovering a univocal reality, applied sciences were reconciled to the possibility of incompatible prescriptions. Following this, we introduce some concepts and approaches central to discussions of agricultural biotechnologies including social epistemology and social ontology. Our coinage, engineered agricultural ecologies is meant to echo debate over novel ecosystems in conservation biology, where attempts to intervene in ecosystems damaged by species loss and climate change are advocated by some, opposed by others. By insisting that farms are engineered ecologies we hope to stimulate some crosstalk with environmental philosophers and conservation ecology. © The Author(s) 2025.